

$$\bullet \text{ res} = x_{\text{mid}} - x_{\text{teo}} \quad , \quad \text{con } x_{\text{teo}} = x_{\text{izq}} + (x_{\text{der}} - x_{\text{izq}}) \cdot \underbrace{(y_{\text{mid}} - y_{\text{izq}}) / (y_{\text{der}} - y_{\text{izq}})}_{\alpha}$$

$$\Rightarrow \text{res} = x_{\text{mid}} - x_{\text{izq}}(1-\alpha) - \alpha x_{\text{der}}$$

$$\begin{aligned} \Rightarrow \text{Var}(\text{res}) &= \sigma_{\text{hit}}^2 + (1-\alpha)^2 \sigma_{\text{hit}}^2 + \alpha^2 \sigma_{\text{hit}}^2 = \quad (\sigma_{\text{hit}} \equiv \text{smearing}) \\ &= \sigma_{\text{hit}}^2 \left(1 + (1-\alpha)^2 + \alpha^2 \right) = \underbrace{\sigma_{\text{hit}}^2}_{m^2} \cdot \underbrace{1 - \left(\frac{y_{\text{mid}} - y_{\text{izq}}}{y_{\text{der}} - y_{\text{izq}}} \right)^2}_{\frac{y_{\text{der}} - y_{\text{mid}}}{y_{\text{der}} - y_{\text{izq}}}} \\ &= \sigma_{\text{hit}}^2 \left(1 + \left(\frac{y_{\text{der}} - y_{\text{mid}}}{y_{\text{der}} - y_{\text{izq}}} \right)^2 + \left(\frac{y_{\text{mid}} - y_{\text{izq}}}{y_{\text{der}} - y_{\text{izq}}} \right)^2 \right) \end{aligned}$$

• Ajuste gaussiana simple σ :

$$\text{res} \equiv \sigma = m \cdot \sigma_{\text{hit}} \Rightarrow \text{Var}(\text{res}) = m^2 \cdot \sigma_{\text{hit}}^2 \Rightarrow \text{La pendiente } m \text{ depende del detector que no ha sido impactado:}$$

$\uparrow$
 $\text{Var} \equiv \sigma^2$

En estos datos, tenemos:  , con $\begin{cases} d = 7 \text{ cm} \\ D = 13 \text{ cm} \end{cases}$. Como $m^2 = 1 + (1-\alpha)^2 + \alpha^2$:

- No hit detector 0: $\alpha = \frac{y_2 - y_1}{y_3 - y_1} = \frac{20 - 7}{27 - 7} = \frac{13}{20} \Rightarrow m = 1,243$

- " " " 1: $\alpha = \frac{y_2 - y_0}{y_3 - y_0} = \frac{20}{27} \Rightarrow m = 1,271$

- " " " 2: $\alpha = \frac{y_1 - y_0}{y_3 - y_0} = \frac{7}{27} \Rightarrow m = 1,271$

$\Rightarrow$ - " " " " : $m = 1,243$

Simétrico $\Rightarrow$